# Flexible-margin Gaussian-copula full random-effects models: likelihood estimation and posterior-based margin selection

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**Abstract**

Full random-effects modelling (FREM) jointly describes parameter and covariate random effects by a multivariate Normal population distribution. We extend FREM by combining Gaussian dependence with flexible marginal distributions defined directly for the natural individual parameters and covariates. The internal `saemix` transformation is retained only as a reversible coordinate map with its exact Jacobian. Because individual parameters are latent, we use a screen-and-refit workflow: a standard fit supplies full individual posterior samples, support-compatible natural-scale families are ranked on independent pools, and a resolved selection starts a fresh fit; unresolved cases retain the standard family. For each fixed specification, population, marginal, dependence and residual parameters are estimated jointly by controlled-Markov likelihood-score stochastic approximation with a finitely learned empirical-information preconditioner. In 400 fixed-family nonlinear mixed-effects datasets, structural bias remained within 0.47%. In 500 joint datasets generated with Gamma clearance and Gamma CRP, the flexible workflow selected Gamma clearance 430 times and conservatively retained lognormal in 37. It improved total and conditional-response likelihood in all 500 datasets, population VPC error in 459 and upper-CRP conditional VPC error in 496. Median conditional VPC log-RMSE was 0.277 versus 1.891 for Gaussian FREM. Thus natural-scale marginal assumptions can be learned separately from fixed-model likelihood estimation and materially affect population and conditional tail prediction.

**Keywords:** nonlinear mixed-effects model; full random-effects model; Gaussian copula; stochastic approximation; marginal likelihood; covariates

# INTRODUCTION

Covariate models describe how patient characteristics contribute to variability in individual model parameters and predictions. Full random-effects modelling (FREM) includes a prespecified set of covariates as additional random effects and estimates their joint covariance with the parameter random effects [1]. Covariate-effect coefficients and unexplained parameter variability can then be calculated for any subset of the included covariates from a single estimation. This avoids repeated estimation of separate reduced covariate models and allows missing covariates to be integrated over in the likelihood [2, 3].

Standard FREM assumes that the combined parameter and covariate random effects are multivariate Normal. For covariates observed with negligible error, this working model can estimate means and covariances accurately under some non-Normal covariate distributions [1]. Recent FREM investigations also show that covariate-model misspecification can affect inclusion and omission bias [4]. The Gaussian working model nevertheless specifies the entire population distribution, including marginal shape and conditional tail behaviour. These features are relevant when the fitted model is used for conditional prediction, simulation of virtual populations, integration over missing covariates, or extrapolation to sparsely represented covariate values.

A copula permits the marginal distributions to be specified separately from their dependence [5, 6]. We extend FREM by combining a Gaussian copula with flexible continuous margins for the natural individual parameters and continuous or ordered categorical margins for covariates. The `saemix` parameter transformation remains a computational coordinate map; it does not determine the marginal family. Thus a Gamma clearance remains Gamma distributed whether the internal coordinate is identity or logarithmic. Standard Normal or lognormal parameter models are retained as special cases. Continuous covariate margins can be screened from their observed values, whereas individual-parameter margins require the response data because those parameters are latent. Their screening therefore uses full individual conditional distributions rather than point summaries.

The proposed workflow therefore separates discrete family selection from continuous likelihood estimation. A standard Normal/lognormal incumbent is fitted first. Full draws from its individual conditional distributions are mapped to the natural parameter scale and used to screen registered families compatible with the declared support by an exact observed-likelihood ratio identity. A resolved selection starts a fresh fit; otherwise the standard family is retained. Family names remain fixed within the final recursion, while all continuous parameters remain free.

The objectives of this work were to define the flexible-margin FREM model and its observed-data likelihood, establish a likelihood-score estimator for any fixed admissible margin specification, develop posterior-based automatic natural-parameter margin selection, derive the corresponding conditional and FFEM representations, and evaluate estimation and conditional prediction in simulation studies.

# METHODS

## Statistical model

### Gaussian FREM

For subject $i$ at design point $x_{ij}$, write the nonlinear mixed-effects observation model as [7]

$$y_{ij} = f(\psi_i, x_{ij}) + g(\psi_i, \sigma, x_{ij})\epsilon_{ij}, \qquad \epsilon_{ij} \sim N(0,1), \tag{1}$$

where $\psi_i$ is the vector of individual model parameters and $\sigma$ contains the residual-error parameters. On the transformed additive scale,

$$\phi_i = h(\psi_i) = C_i\mu + \eta_{\mathrm{par},i}, \tag{2}$$

where $C_i$ is the fixed-effect design matrix, $\mu$ is the vector of population parameters, and $\eta_{\mathrm{par},i}$ is the parameter random effect. The lowercase symbol $c_i$ denotes the FREM covariate vector and distinguishes it from the design matrix $C_i$. An error-free additive FREM covariate model is

$$c_i = \mu_c + \eta_{\mathrm{cov},i}. \tag{3}$$

Here $\eta_{\mathrm{par},i}$ represents total parameter variability (TPV) and $\eta_{\mathrm{cov},i}$ is the individual covariate deviation. Standard FREM specifies

$$\begin{pmatrix} \eta_{\mathrm{par},i} \\ \eta_{\mathrm{cov},i} \end{pmatrix} \sim N\left[0, \Omega_{\mathrm{FREM}}\right], \qquad \Omega_{\mathrm{FREM}} = \begin{pmatrix} \Omega_{\mathrm{par}} & \Omega_{\mathrm{par,cov}} \\ \Omega_{\mathrm{cov,par}} & \Omega_{\mathrm{cov}} \end{pmatrix}. \tag{4}$$

Equation (4) is the standard FREM specification [8, 1]. Let $\Omega_{\mathrm{par,cov}} \in \mathbb{R}^{p\times q}$ and $\Omega_{\mathrm{cov,par}} = \Omega_{\mathrm{par,cov}}^{\mathsf{T}}$. The covariate-effect matrix and unexplained parameter variability (UPV) are

$$B = \Omega_{\mathrm{par,cov}}\Omega_{\mathrm{cov}}^{-1}, \qquad \Omega'_{\mathrm{par}} = \Omega_{\mathrm{par}} - B\Omega_{\mathrm{cov,par}}. \tag{5}$$

Consequently,

$$\eta_{\mathrm{par},i} \mid c_i \sim N\left[B(c_i - \mu_c), \Omega'_{\mathrm{par}}\right]. \tag{6}$$

Equations (4)–(6) give the standard Gaussian FREM model, its conditional covariate coefficients, and its UPV representation.

**Flexible-margin Gaussian-copula FREM**

Let $X_i = (\psi_i, c_i)$ contain the natural individual parameters and covariates. For continuous coordinate $j$, write $F_j(\cdot;\alpha_j)$ and $f_j(\cdot;\alpha_j)$ for its natural-scale CDF and density and define

$$\xi_j = \Phi^{-1}\{F_j(x_j;\alpha_j)\}.$$

Let $R(\rho)$ be a positive-definite Gaussian-score correlation matrix and let $\theta_p = (\alpha, \rho)$ collect the population-distribution parameters. The continuous population density is

$$p_{\theta_p}(x) = |R|^{-1/2} \exp\left[-\tfrac{1}{2}\xi^{\mathsf{T}}(R^{-1} - I)\xi\right] \prod_{j=1}^{d} f_j(x_j;\alpha_j). \tag{7}$$

The model transformation $\phi = h(\psi)$ is not part of the marginal-family definition. When computation uses $\phi$, the equivalent density is

$$p_\phi(\phi) = p_\psi\{h^{-1}(\phi)\}\,|J_{h^{-1}}(\phi)|\,, \tag{7b}$$

so identity and logarithmic storage give the same natural-parameter likelihood. For example, a Gamma clearance remains Gamma; it is not a Gamma random effect followed by exponentiation.

We parameterize $R$ by a fixed regular vine with Gaussian pair copulas; the vine structure is not estimated during the stochastic-approximation recursion. When the natural margins induce Normal additive effects under the declared maps (Normal under identity and lognormal under a log map), (7) reduces exactly to the error-free Gaussian FREM population model (2)–(4), with $\Omega_{\mathrm{FREM}}$ obtained from the marginal standard deviations and $R$. This error-free formulation is the zero-covariate-residual limit of the conventional NONMEM/PsN formulation, which uses a small fixed residual variance for covariate responses [1, 3].

Let $\theta = (\mu, \alpha, \rho, \sigma)$ denote the unknown parameter vector. Here $\alpha$ contains the parameters of the marginal distributions of $\psi$ and $c$, including their identified location anchors; $\rho$ parameterizes $R$; and $\sigma$ contains the residual-error parameters. When the declared maps yield Gaussian additive parameter and covariate effects, the marginal scales together with $\rho$ are equivalently represented by $\mathrm{vec}(\Omega_{\mathrm{FREM}})$; the covariate locations $\mu_c$ remain separate.

The corresponding FREM effect summaries live naturally on the latent-score scale. Partition $R$ into individual-parameter and covariate blocks. Then

$$B_\xi = R_{\mathrm{par,cov}} R_{\mathrm{cov}}^{-1}, \qquad R'_{\mathrm{par}} = R_{\mathrm{par}} - B_\xi R_{\mathrm{cov,par}}, \tag{7a}$$

so that $\xi_{\mathrm{par}} \mid c$ is Normal with mean $B_\xi \xi_{\mathrm{cov}}(c)$ and covariance $R'_{\mathrm{par}}$. The fitted marginal quantile maps transform this conditional law back to the native parameter scale. Under all-Normal margins, let $D_{\mathrm{par}}$ and $D_{\mathrm{cov}}$ be diagonal matrices of marginal standard deviations. Then $B = D_{\mathrm{par}} B_\xi D_{\mathrm{cov}}^{-1}$ and $\Omega'_{\mathrm{par}} = D_{\mathrm{par}} R'_{\mathrm{par}} D_{\mathrm{par}}$, which recovers (5). Under non-Gaussian parameter margins the native-scale conditional mean need not be linear. In that case, $B_\xi$, $R'_{\mathrm{par}}$ and the marginal quantile functions jointly determine the conditional distribution.

## Estimation and margin-selection workflow

The complete workflow contains four steps. First, continuous covariate margins are either prespecified or screened from their observed values; categorical margins and their ordering follow their scientific meaning. Second, a standard Normal/lognormal Gaussian-copula incumbent is fitted. Third, full draws from its individual conditional distributions are mapped to the natural parameter scale and used to rank registered families compatible with each declared support. Fourth, a resolved family combination starts a fresh fixed-family fit that re-estimates all population, dependence and residual-error parameters. An unresolved screen is repeated with a larger posterior pool and otherwise retains the standard model. Users may restrict screening to selected parameters or bypass it by prespecifying scientifically justified margins.

This separation is essential: the fixed-family estimator has one unchanged state space and likelihood throughout its recursion, while family selection is a finite outer model-comparison step. The convergence result below concerns the incumbent and final fixed-family fits, not the discrete selection operation.

**Observed-data likelihood**

Let $O_i = (Y_i, c_{i,\mathrm{obs}})$ and let $Z_i$ collect individual parameters and missing covariates. For categorical covariates and continuous margins with parameter-dependent support, $Z_i$ additionally contains the fixed-support unit-interval variables defined below. With a fixed mixed Lebesgue/counting measure $\lambda$, define the complete-data likelihood $f_i$ and observed-data likelihood $g_i$ by

$$g_i(\theta) = \int f_i(z_i;\theta)\,\lambda(dz_i), \qquad \ell(\theta) = \sum_{i=1}^{N} \log g_i(\theta). \tag{8}$$

The dependence of $f_i$ and $g_i$ on $O_i$ is implicit, as in Delyon et al.; $Z_i$ denotes the missing or augmented data [9]. Consequently, the likelihood contribution is based on the joint density $p_\theta(Y_i, c_{i,\mathrm{obs}})$. Individual effects and missing covariates are integrated out rather than replaced by point estimates. The analysis conditions on the recorded missingness pattern and assumes that the covariate-missingness mechanism is ignorable.

When all covariates are observed, its subject contribution has the exact factorization

$$\log p_\theta(Y_i, c_i) = \log p_{\theta_p}(c_i) + \log p_\theta(Y_i \mid c_i). \tag{8a}$$

Thus a difference in joint log likelihood may reflect differences in the fitted covariate density, the conditional response likelihood, or both. This factorization does not alter the estimation objective in (8).

When the continuous margins are proper and absolutely continuous, the required fixed-support representations exist, the categorical probability vectors are proper, $R$ is positive definite, and the response density is proper, (7) defines a probability distribution and $g_i(\theta)$ in (8) is a valid observed-data likelihood contribution. This follows from a componentwise probability-integral transformation: continuous Jacobians transform (7) to a $N_d(0, R)$ density, categorical cells partition their latent Gaussian coordinates, and the fixed-support representations put moving boundaries on fixed unit intervals. Appendix A gives the detailed normalization argument.

**Fixed-family estimation problem**

For the Gaussian FREM model, the complete-data likelihood has the curved exponential-family representation used by standard SAEM,

$$\log p_\theta(Y, Z) = -\psi(\theta) + \langle S(Y, Z), \varphi(\theta)\rangle,$$

where $S(Y, Z)$ is a parameter-independent finite-dimensional sufficient statistic. Standard SAEM recursively averages $S(Y, Z)$ and performs the M-step from that finite vector [9, 10].

In (7), a marginal parameter $\alpha_j$ enters both $\log f_j(x_j; \alpha_j)$ and the transformed value $\xi_j = \Phi^{-1}\{F_j(x_j; \alpha_j)\}$ inside the copula quadratic form. Marginal and dependence parameters are therefore coupled through nonlinear functions of the latent data, and a parameter-independent finite statistic $S(Y, Z)$ is not generally available. The observed-data likelihood and the EM identity remain valid, but the usual finite-statistic SAEM recursion and its convergence theorem cannot simply be carried over; the curved-exponential condition is a recognized bottleneck for SAEM [11]. One possible workaround adds fixed-variance Gaussian latent variables to exponentialize the model, but then maximizes a perturbed likelihood whose approximation and numerical behaviour depend on that variance [12, 11]. Here the likelihood score of the original, unperturbed model is used instead, through the stochastic-gradient recursion proposed by Younes and presented as equation (74) by Delyon et al. [13, 9]. When the declared maps yield Gaussian additive effects, the ordinary Gaussian FREM finite-statistic representation is recovered.

### Controlled-Markov approximation of the likelihood score

Work in unconstrained internal coordinates $\vartheta$, with native parameters $\theta = \tau(\vartheta)$. Log, logit and hyperspherical transformations enforce positive scales, probabilities and a positive-definite $R$. Write $\widetilde{f}_i(z;\vartheta) = f_i\{z;\tau(\vartheta)\}$ and $\widetilde{g}_i(\vartheta) = g_i\{\tau(\vartheta)\}$. At iteration $k$, controlled Markov kernels invariant for the current individual conditional law advance $m$ chains per subject. The estimated complete score is

$$\widehat{H}_{k+1} = \frac{1}{mN}\sum_{r=1}^{m}\sum_{i=1}^{N}\nabla_\vartheta \log f_i(Z_{i,k+1}^{(r)};\tau(\vartheta_k)), \tag{9}$$

and the parameter update is

$$\vartheta_{k+1} = \mathrm{Proj}_K\{\vartheta_k + \gamma_{k+1}A_k\widehat{H}_{k+1}\}. \tag{10}$$

During a finite learning sequence, the diagonal empirical score-information estimate is updated from squared stochastic-score coordinates. The preconditioner is $A_k = \{\mathrm{diag}(\widehat{I}_k)+\lambda I\}^{-1}$ with bounded eigenvalues; it replaces a family-specific manual score multiplier and approximates inverse-information scaling. Both $A_k$ and proposal scales are subsequently held fixed. The gain sequence is $\gamma_k = \gamma_0(k+k_0)^{-a}$ with $3/4 < a \leq 1$. Marginal families, category order, copula structure and all other discrete model choices are specified before recursion (10) begins.

Equation (10) is a controlled-Markov, internal-coordinate, projected, preconditioned and multiple-chain extension of the Younes stochastic-gradient recursion presented as Delyon et al. equation (74), $\theta_k = \theta_{k-1} + \gamma_k\partial_\theta \log f(Z_k;\theta_{k-1})$. The coordinate transformations and fixed positive-definite metric leave the stationary points unchanged. Because the implemented chains persist between iterations, their draws are Markovian and do not satisfy the conditional-independence assumption used in Delyon et al.'s direct analysis of equation (74). Appendix B replaces that martingale-difference argument by the Poisson-equation decomposition for controlled-Markov stochastic approximation, building on the SAEM–MCMC construction and its later extensions [14, 15].

Every stochastic move is either an exact conditional draw or a Metropolis- Hastings transition with the current individual conditional law as invariant distribution. At least one exact conditional-population independence proposal is used per iteration. With several categorical covariates, random-walk moves that would require a numerical rectangle probability are omitted; the exact independence kernel remains. An unadjusted approximate kernel would generally have a different invariant distribution and hence a different mean field [16].

### Likelihood target

Provided differentiation can pass through (8), Fisher's identity gives

$$E_\vartheta\{\nabla_\vartheta \log \widetilde{f}_i(Z_i;\vartheta) \mid O_i\} = \nabla_\vartheta \log \widetilde{g}_i(\vartheta). \tag{11}$$

For one subject, substitution of the conditional density $\widetilde{f}_i(z;\vartheta)/\widetilde{g}_i(\vartheta)$ into the expectation reduces its numerator to $\nabla_\vartheta \int \widetilde{f}_i(z;\vartheta)\lambda(dz)$. The fixed-support representations ensure that $\lambda$ is fixed. Appendix A states the required domination conditions and gives the full calculation [17].

Let $\bar{\ell} = N^{-1}\ell$. Thus the mean field after finite tuning is $A\nabla\bar{\ell}$. For symmetric positive-definite $A$,

$$\frac{d}{dt}\bar{\ell}\{\vartheta(t)\} = \nabla\bar{\ell}(\vartheta)^{\mathsf{T}}A\nabla\bar{\ell}(\vartheta) \geq 0. \tag{12}$$

### Convergence

Let $\mathcal{L} = \{\vartheta : \nabla\bar{\ell}(\vartheta) = 0\}$. Suppose the fixed model is continuously differentiable on the stabilized compact interior set; differentiation is dominated; the controlled kernels have unique invariant laws, uniform geometric drift/minorization and the local Poisson-equation regularity; the gain satisfies the Fort condition; adaptation is finite and numerical error is gain-weighted summable; safeguards eventually cease; and $-\bar{\ell}$ has the required coercive Lyapunov extension and critical-value property. Then controlled-Markov stochastic approximation gives

$$d(\vartheta_k, \mathcal{L}) \longrightarrow 0 \qquad \text{almost surely.} \tag{13}$$

The limit set lies in a connected component of $\mathcal{L}$ and collapses to an isolated attracting point when one contains the tail. This is a conditional stationary-set result, not a global-maximum claim and not an application of the finite-statistic SAEM Theorems 5 or 8. Appendix B states and maps the complete assumptions [18, 15].

### Posterior-based automatic natural-margin selection

Marginal families are discrete model choices and are not changed during recursion (10). They are screened before a fresh fixed-model fit by reusing the complete individual posterior distribution from a standard incumbent. Let $f_{0i}$ and $g_{0i}$ denote the incumbent complete- and observed-data likelihoods, and let $\pi_{0i}(z) = f_{0i}(z)/g_{0i}$ be its individual conditional distribution. For a candidate model $m$ with the same observation model, population locations, covariate margins and Gaussian dependence during screening,

$$\frac{g_{mi}}{g_{0i}} = E_{\pi_{0i}}\left\{\frac{f_{mi}(Z_i)}{f_{0i}(Z_i)}\right\} = E_{\pi_{0i}}\left\{\frac{p_m(\psi_i, c_i)}{p_0(\psi_i, c_i)}\right\}. \tag{14}$$

Unlike EBEs, these draws retain individual uncertainty. Candidate parameters are optimized on one posterior pool; an independently generated pool gives the validation BIC advantage

$$2\sum_{i=1}^{N}\log\left[\frac{1}{M}\sum_{r=1}^{M}\frac{p_m(\psi_i^{(r)}, c_i)}{p_0(\psi_i^{(r)}, c_i)}\right] - (k_m - k_0)\log N, \tag{15}$$

relative to the incumbent. Candidate families are supplied by a support-based registry; the present implementation includes Normal, Student and Laplace on the real line and lognormal, Gamma and Weibull on the positive line. The model transformation does not restrict this registry. Incumbent dependence and non-screened margins are held fixed during ranking. MCMC ESS, importance ESS, batch-means Monte Carlo error and ranking separation determine whether the result is resolved. The log of the importance average has conservative Jensen bias toward the incumbent, and Normal-to-Student weights can have infinite variance; empirical ESS detects concentration but cannot prove a finite second moment. Unresolved screens require more draws, fewer candidates or direct comparison of refitted leaders.

The selected family names are held fixed for a fresh score fit that re-estimates all continuous population, dependence and residual-error parameters. Thus (13) applies to that fixed-model fit, not to selection itself. Appendix A proves (14) and gives the estimator qualifications.

## Conditional representation and FFEM translation

For a selected covariate subset $S$, let $z_{iS}$ contain the corresponding Gaussian scores and calculate $B_{\xi,S}$ and $R'_{\mathrm{par},S}$ from the relevant blocks of $R$. If the transformed parameter coordinates are Gaussian, the exact FFEM is

$$\phi_i = C_i\mu + \beta_{z,S}z_{iS} + \eta'_{iS}, \qquad \beta_{z,S} = D_{\mathrm{par}}B_{\xi,S}, \qquad \eta'_{iS} \sim N(0, \Omega'_{\mathrm{par},S}),$$

where $\Omega'_{\text{par},S} = D_{\text{par}} R'_{\text{par},S} D_{\text{par}}$. Covariates are fixed inputs whose fitted CDFs produce $z_{iS}$; all-Normal covariate margins recover the usual raw-scale FREM formulas [1].

For other natural-parameter margins, a conventional additive-Normal FFEM is not generally equivalent. Let $Q_\psi$ apply the fitted natural-parameter quantile functions componentwise and choose $L_S$ such that $L_S L_S^\intercal = R'_{\text{par},S}$. The exact conditional representation is

$$\psi_i = Q_\psi[\Phi\{B_{\xi,S} z_{iS} + L_S \varepsilon_i\}], \qquad \varepsilon_i \sim N(0, I).$$

This transformed latent-Normal law retains the non-Normal margins exactly; it is a conditional generative representation, not a conventional additive-Normal FFEM. Ordered categories use the corresponding latent-interval mixture. The translation represents $p(y \mid c_S)$ and therefore omits the covariate-density term in the joint FREM likelihood. Appendix A proves both cases.

## Categorical covariates and parameter-dependent support

Binary, ordinal and explicitly ordered nominal covariates are represented by their latent Gaussian intervals; continuous moving-support coordinates use a fixed percentile and the candidate quantile map. Both augmentations live on a parameter-independent reference space, so Fisher's identity remains applicable and multiple categorical coordinates are allowed. Permutation-invariant nominal models and nonregular support-boundary MLEs remain outside scope. Appendix A gives the constructions and normalization.

## Simulation and numerical evaluation

### Fixed-family re-estimation

The primary simulation isolated fixed-family estimation from family selection. Four scenarios (Normal/Normal, lognormal/Normal, Gamma/Normal and Weibull/lognormal covariates) each contained 100 independently generated one-compartment datasets with $N = 100$, two eta coordinates and two covariates. Generating families were supplied to the flexible fit. Each fit used 1,500 score iterations with persistent invariant kernels and gain exponent 0.80; all adaptation stopped after 50 iterations. Gaussian and flexible fits were reevaluated with independent 5,000-draw importance samples. The complete parameters, dependence matrix, seeds and stabilization criteria are recorded in the reproducibility bundle.

### Joint natural-parameter and covariate-margin study

Five hundred independent one-compartment datasets contained $N = 250$ subjects with eight observations each. Natural volume was lognormal and natural clearance was Gamma with shape 2.5 and geometric mean 3.5. CRP was Gamma with shape 2 and scale 20. Their latent Gaussian-score correlations were 0.20 for volume–clearance, 0.15 for volume–CRP and 0.60 for clearance–CRP; proportional residual SD was 0.12. The Gaussian FREM comparator used Normal parameter random effects and a Normal CRP margin. For the flexible workflow, the CRP family was screened from its observed values, a fixed flexible-covariate incumbent supplied full individual posterior draws, and positive clearance candidates (lognormal, Gamma and Weibull) were fitted on 700 draws per subject and ranked on an independent 700-draw pool. An unresolved ranking was repeated with 2,000 draws; persistent uncertainty retained lognormal clearance. Every fitted model used 1,500 score iterations, and a resolved selection was fitted afresh with new SA and MCMC state. Independent 2,500-draw importance samples evaluated the joint observed likelihood and its covariate and conditional-response components.

For each fitted population, 50,000 common-random-number subjects supplied the natural clearance density and population response percentiles at the 1st, 10th, 50th, 90th and 99th levels. Conditional-median clearance was evaluated over the 1st–99.9th percentiles of the generating CRP distribution. A second population of 50,000 subjects sampled above the 95th CRP percentile supplied conditional 10th, 50th and 90th response percentiles. Population and conditional predictive errors were log-RMSE against generating quantiles.

### Numerical verification

Analytical complete-data scores, Fisher's identity, categorical augmentation, moving-support paths, the Gaussian nested model and the FREM-to-FFEM mapping were checked against independent calculations. Because Gamma shape and scale can compensate while retaining the marginal mean, a separate direct-MLE experiment used 10,000 independently generated Gamma samples at each of $N = 50$, 100, 300, 1,000 and 3,000. Full-model Gamma estimates were also compared with direct Gamma MLEs on the same $N = 100$ and $N = 300$ datasets.

# RESULTS

## Fixed-family re-estimation

Table 1: Flexible-margin FREM parameter recovery and difference in estimated joint observed-data log likelihood relative to Gaussian FREM. Bias is relative mean bias across 100 datasets; $\Delta\ell$ is flexible-margin minus Gaussian FREM.

| Scenario | $V$ bias (%) | $CL$ bias (%) | Residual bias (%) | $\Delta\ell$ |
|---|---|---|---|---|
| Normal/Normal | 0.29 | -0.03 | 0.03 | -0.06 |
| lognormal/Normal | -0.03 | -0.46 | -0.15 | 4.30 |
| Gamma/Normal | 0.31 | 0.18 | -0.41 | 4.60 |
| Weibull/lognormal | -0.26 | 0.01 | 0.05 | 9.41 |

Structural RMSE was nearly identical between methods. Marginal parameters were close to generating values: estimated lognormal sdlog values were 0.239 and 0.277 (truth 0.240 and 0.280), and Weibull shape/scale were 2.309/78.11 (truth 2.3/78). Dataset-specific Gamma estimates matched direct MLEs within mean absolute differences 0.0065 (shape) and 0.0101 (scale). Maximum absolute mean correlation bias was 0.022.

Flexible margins had larger observed-data likelihood in 285/300 non-Gaussian datasets. Most mean improvement arose from the covariate-density component (4.01, 4.15 and 8.58), with conditional-response contributions 0.29, 0.45 and 0.83. The models are non-nested and the independent importance-sampling comparisons are descriptive; by (8a), joint improvement alone does not establish improved $p(Y \mid c)$.

## Joint natural-parameter and covariate-margin study

The flexible workflow selected Gamma clearance in 430 datasets, Weibull in 33 and lognormal in 37 conservative retentions. CRP was selected as Gamma in 389, Weibull in 103 and lognormal in 8. Total and conditional-response likelihoods were higher in all 500 paired datasets and exceeded twice combined Monte Carlo error in all 500. Median gains were 63.23 and 17.33, respectively.

Median population VPC log-RMSE was 0.202 for flexible versus 0.853 for Gaussian FREM, with the flexible fit better in 459/500 datasets. Median upper-CRP conditional VPC error was 0.277 versus 1.891, with improvement in 496/500; the conditional clearance relationship improved in 499/500 (median log-RMSE 0.050 versus 0.167; Figure 1). Mean flexible estimates were 20.05 for $V$, 3.54 for $CL$ and 0.122 for residual SD (generating values 20, 3.5 and 0.12).

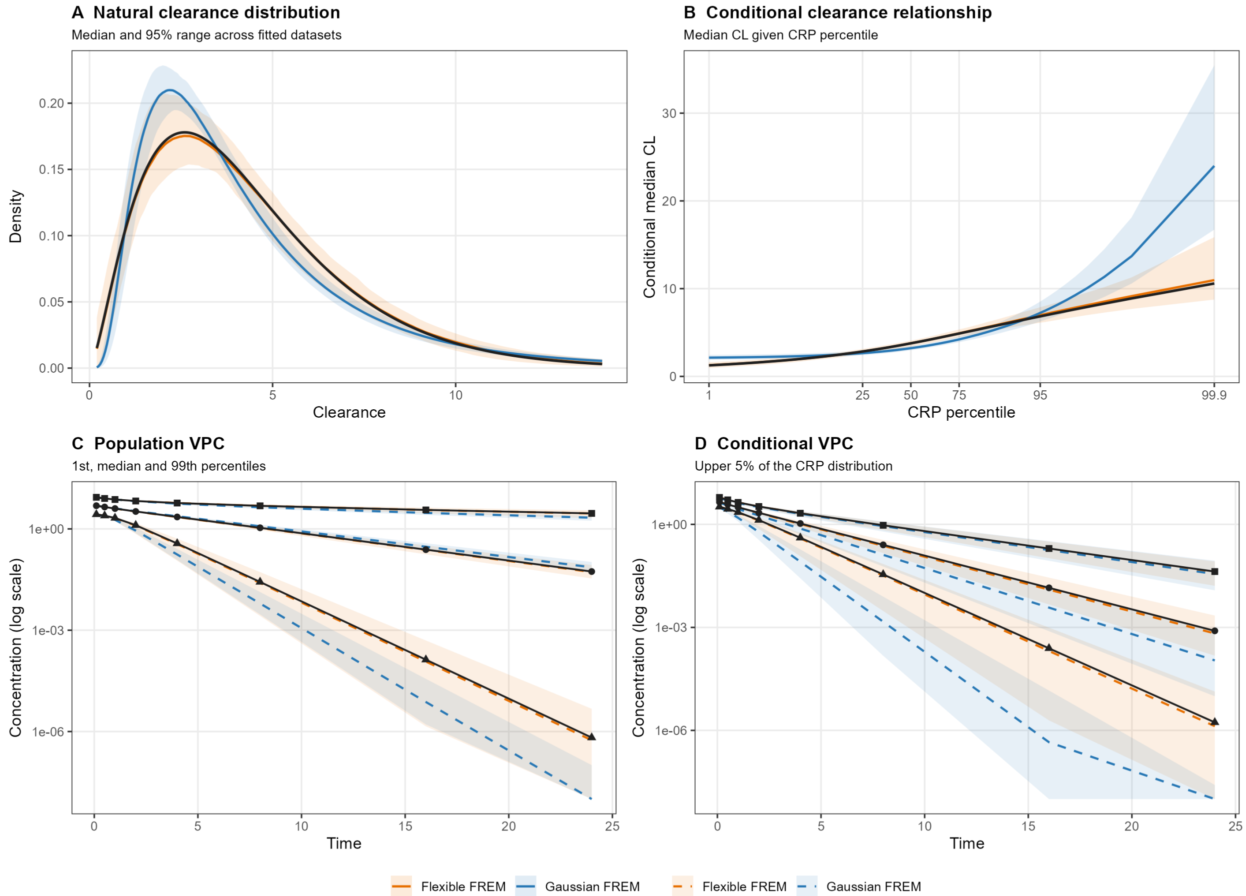


Figure 1: Joint natural-parameter and covariate-margin validation across 500 independent datasets. Black lines and symbols denote the generating model; coloured lines are fitted medians and ribbons are pointwise 95% ranges. (A) Natural clearance density. (B) Conditional median clearance across CRP percentiles. (C) Population VPC at the 1st, 50th and 99th percentiles. (D) Conditional VPC above the 95th CRP percentile at the 10th, 50th and 90th percentiles.

## Numerical verification

Analytical scores agreed with independent derivatives within $6.5 \times 10^{-8}$; Fisher discrepancies were below their Monte Carlo error, including multiple categorical and moving-support examples. Gamma shape bias decreased from 2.83% at $N = 100$ to 0.14% at $N = 3,000$, while full-model and same-dataset direct MLEs agreed within 0.0075 at $N = 300$. Gaussian FREM block formulas, non-Normal conditional draws and direct-versus-translated FFEM predictions agreed to the stated numerical tolerances in the reproducibility tests.

# DISCUSSION

The method separates the distribution assigned to each natural individual parameter from the coordinate map used by the nonlinear mixed-effects software. A standard Normal or lognormal model supplies full posterior draws; registered support-compatible families are ranked outside the recursion, and a resolved selection is fitted afresh. Each fit therefore retains one likelihood and one state space. In the 500-dataset joint study, this workflow usually selected Gamma clearance and a non-Gaussian CRP margin, while unresolved comparisons conservatively retained lognormal clearance.

The clearest practical difference was prediction conditional on covariate-tail values. A misspecified marginal CDF changes the Gaussian score assigned to a covariate value and therefore changes the fitted distribution $p(\psi \mid c)$ even when the first two moments are estimated accurately. Across 500 fits, flexible margins substantially reduced conditional VPC-quantile error and contained the generating upper-tail relationship within their between-fit band.

The conditional representation is a conventional transformed-covariate FFEM for Normal parameter margins and an exact quantile-transformed generative model otherwise. Direct and translated predictions coincide at fixed population parameters; re-estimation of the translation instead targets $p(y \mid c_S)$.

The controlled-Markov score recursion changes the computation, not the target: Fisher's identity gives the observed-likelihood gradient. The diagonal empirical-information metric is adapted only during a finite initial period and then held fixed, so it changes numerical scaling but not the mean-field zeros. Convergence remains conditional on Appendix B's assumptions. Automatic screening is likewise diagnosed rather than unconditional: Jensen bias favours the incumbent, and heavy-tailed candidates may require more draws, defensive proposals or direct refitted-likelihood comparison despite an apparently adequate finite ESS.

The evaluation used one PK model. Broader structural models and formal coverage studies remain to be evaluated. General non-Gaussian R-vines, nonregular support-boundary MLEs, permutation-invariant nominal models and fixed-only estimated structural parameters are outside scope. Supported margins must be proper, identifiable and satisfy the stated differentiability and moment conditions; this is not a distribution-free estimator.

# CONCLUSION

Flexible-margin Gaussian-copula FREM separates natural-parameter and covariate margins from dependence and contains Gaussian FREM when the declared maps yield Gaussian additive effects. Natural-parameter families need not be known in advance: a standard incumbent, full-posterior screen, diagnosed independent validation and fresh selected-family fit form one coherent likelihood workflow. For every fixed family specification, Fisher's identity links the controlled-Markov stochastic-approximation mean field to the observed-data likelihood score. In the simulations considered here, structural-parameter recovery remained accurate, while flexible margins improved the observed likelihood and population or conditional tail prediction under non-Gaussian generating distributions. The fitted conditional model translates exactly to a conventional FFEM when transformed parameter coordinates are Gaussian and to an exact conditional generative representation otherwise.

# DECLARATIONS

Funding, conflicts of interest, author contributions, data availability and software availability will be completed before submission.

# APPENDIX A: Likelihood normalization, Fisher identity and FFEM translation

## Continuous normalization

For fixed $\theta$, the map $x_j \mapsto \xi_j = \Phi^{-1}\{F_j(x_j)\}$ has Jacobian $d\xi_j/dx_j = f_j(x_j)/\phi(\xi_j)$. Substitution in (7) therefore gives

$$p_{\theta_p}(x)\, dx = \phi_R(\xi)\, d\xi.$$

Integration over $\mathbb{R}^d$ is one. This proves normalization for any proper absolutely continuous margins; Gaussian marginal shape is not required.

## Several categorical covariates

For observed category vector $k_{\mathcal{D}}$, let $u_j = b_{j,k_j-1} + v_j\pi_{j,k_j}$ for $j \in \mathcal{D}$. The augmented mixed density is

$$p_\theta^{\text{aug}}(x_{\mathcal{C}}, k_{\mathcal{D}}, v_{\mathcal{D}}) = c_R(u) \prod_{j\in\mathcal{C}} f_j(x_j) \prod_{j\in\mathcal{D}} \pi_{j,k_j}. \tag{A1}$$

Since $du_j = \pi_{j,k_j} dv_j$, integrating $v_{\mathcal{D}} \in (0,1)^{|\mathcal{D}|}$ gives the Gaussian-copula rectangle mass for $k_{\mathcal{D}}$. Summing category vectors tiles the latent unit cube, so the masses sum to one. Missing categories add counting-measure summation and remain normalized.

## Moving continuous support

For $X = F_\alpha^{-1}(W)$ with $W \in (0,1)$, $dW = f_\alpha(X)dX$. Thus a latent moving-support coordinate contributes the fixed reference $dW$ and its marginal density factor disappears after change of variables. Dependence still enters through $\xi = \Phi^{-1}(W)$. If the coordinate affects the response, $p_\theta\{Y \mid h^{-1}(C\mu + F_\alpha^{-1}(W))\}$ is differentiated through both $\mu$ and $F_\alpha^{-1}$. The moving-boundary derivative is therefore represented by the pathwise quantile derivative rather than discarded.

## Observed likelihood and Fisher identity

Multiplying the normalized population density by the conditional response density and integrating over all unobserved coordinates gives $g_i(\theta)$. Under an integrable derivative envelope,

$$\begin{aligned} E_\vartheta[\nabla \log \widetilde{f}_i(Z_i;\vartheta) \mid O_i] &= \int \frac{\nabla \widetilde{f}_i(z;\vartheta)}{\widetilde{f}_i(z;\vartheta)} \frac{\widetilde{f}_i(z;\vartheta)}{\widetilde{g}_i(\vartheta)}\, \lambda(dz) \\ &= \frac{\nabla \int \widetilde{f}_i(z;\vartheta)\lambda(dz)}{\widetilde{g}_i(\vartheta)} = \nabla \log \widetilde{g}_i(\vartheta). \end{aligned}$$

The chain rule gives $\nabla_\vartheta \bar{\ell} = J_\tau(\vartheta)^\mathsf{T} \nabla_\theta \bar{\ell}$. A nonsingular $J_\tau$ makes native- and internal- coordinate stationary points equivalent.

## Gaussian nested model and FFEM translation

When every margin is Normal, $X = \mu + D\xi$ with $\xi \sim N_d(0,R)$, hence $X \sim N_d(\mu, DRD)$ and $\Omega_{\text{FREM}} = DRD$ exactly.

Let $P$ index the parameter random effects and let $S$ be any selected subset of continuous covariates. Marginalizing covariates outside $S$ leaves the Gaussian subvector $(\xi_P, \xi_S)$ with the corresponding submatrix of $R$. The multivariate-Normal conditioning formula gives

$$\xi_P \mid \xi_S = z_S = B_{\xi,S} z_S + L_S \varepsilon, \qquad B_{\xi,S} = R_{P,S} R_{S,S}^{-1}, \qquad L_S L_S^\mathsf{T} = R_{P,P} - B_{\xi,S} R_{S,P}, \tag{A2}$$

where $\varepsilon \sim N(0,I)$. If the transformed parameter coordinates are Gaussian, $\phi_P = C_P\mu + D_P\xi_P$; substitution in (A2) yields the coefficient matrix $D_P B_{\xi,S}$ and UPV covariance $D_P L_S L_S^\mathsf{T} D_P$ stated in the conventional FFEM representation. If the selected covariate margins are also Normal, replacing $z_S$ by $D_S^{-1}(c_S - \mu_S)$ gives the usual raw-scale FREM coefficient matrix and Schur-complement UPV covariance.

For other continuous natural-parameter margins, let $Q_\psi$ apply their quantile functions componentwise. Since $\psi_P = Q_\psi\{\Phi(\xi_P)\}$, (A2) gives

$$\psi_P \mid c_S = Q_\psi\left[\Phi\left\{B_{\xi,S} z_S(c_S) + L_S\varepsilon\right\}\right]. \tag{A3}$$

Equation (A3) is the exact non-Normal conditional representation in the main text and has the same conditional law as the fitted FREM. For an ordered categorical covariate, conditioning is instead on its latent Gaussian interval; integrating (A2) over the corresponding truncated-Normal score distribution gives the stated interval mixture. Finally, direct FREM and translated-FFEM predictions integrate the same response kernel $p(Y \mid \psi_P)$ against this same conditional distribution. Their conditional predictive distributions are therefore identical when the population parameters are held fixed.

## Posterior likelihood-ratio identity

Let $f_{0i}$ and $f_{mi}$ be incumbent and candidate complete-data densities with respect to the same fixed measure $\lambda$, and assume $f_{mi} = 0$ wherever the ratio is otherwise undefined. Since $\pi_{0i} = f_{0i}/g_{0i}$,

$$\begin{aligned} E_{\pi_{0i}}\left(\frac{f_{mi}}{f_{0i}}\right) &= \int \frac{f_{mi}(z)}{f_{0i}(z)} \frac{f_{0i}(z)}{g_{0i}} \lambda(dz) \\ &= \frac{1}{g_{0i}} \int f_{mi}(z)\, \lambda(dz) = \frac{g_{mi}}{g_{0i}}. \end{aligned} \tag{A4}$$

When only the natural-parameter distribution changes, the response density and coordinate-map Jacobian cancel, giving (14). The identity requires candidate and incumbent laws to have compatible support and the candidate to be absolutely continuous with respect to the incumbent. It does not by itself ensure finite variance or adequate mixing; Student ratios may have infinite second moments. Independent fitting and validation pools separate optimization noise, after which the pools are discarded and the selected model is fitted afresh.

# APPENDIX B: Controlled kernel, convergence and numerical error

## Exact conditional-population proposal

Let $q_{\vartheta,i}(\psi)$ denote the population density of $\psi_i$ conditional on observed covariates, from which proposals are sampled exactly, and let $L_{\vartheta,i}(\psi) = p_\theta(Y_i \mid \psi, x_i)$. Then

$$\pi_{\vartheta,i}(\psi) = \frac{L_{\vartheta,i}(\psi) q_{\vartheta,i}(\psi)}{\int L_{\vartheta,i}(v) q_{\vartheta,i}(v) dv}.$$

Suppose uniformly on the compact parameter set that $L_{\vartheta,i}(\psi) \le M_i$ and its normalizer is at least $m_i > 0$. The independence-Metropolis weight $w = \pi/q$ is bounded by $M_i/m_i$, and therefore

$$P^{\mathrm{ind}}_{\vartheta,i}(\psi, A) \ge (m_i/M_i)\Pi_{\vartheta,i}(A). \tag{B1}$$

Composition with any other invariant Metropolis kernels preserves the minorization. With finitely many subjects and chains, the product chain has a uniform Doeblin bound. Exact conditional augmentation after the eta move also preserves it: if $R_\vartheta(da \mid \psi')$ draws missing/categorical/fixed- reference variables, then

$$\widetilde{P}_\vartheta\{(\psi, a), d\psi' da'\} = P_\vartheta(\psi, d\psi') R_\vartheta(da' \mid \psi') \ge \epsilon \Pi_\vartheta(d\psi', da'). \tag{B2}$$

If a rejection sampler for a multivariate category rectangle reaches its finite iteration limit, the iteration is terminated without substituting an approximate draw.

## Markovian stochastic-approximation decomposition

After finite adaptation, write the score noise as the conditional mean plus a martingale term, a Poisson-equation remainder and numerical error:

$$\widehat{H}_{k+1} = \nabla \bar{\ell}(\vartheta_k) + e_{k+1} + r^{\mathrm{P}}_{k+1} + r^{\mathrm{num}}_{k+1}. \tag{B3}$$

Here is the precise mapping to controlled-Markov stochastic approximation [19, 20, 14, 15]. On every visited compact parameter set, assume their H1–H2 with a drift function $W$: the partially marginalized score is uniformly $W$-bounded, each $P_\vartheta$ has unique invariant distribution $\Pi_\vartheta$, the kernels are uniformly $W$-geometrically ergodic, and, for some $p \ge 2$,

$$\sup_\vartheta P_\vartheta W^p \le \rho W^p + b, \qquad \rho < 1. \tag{B4}$$

The common minorization in (B1)–(B2), together with Lipschitz dependence of the kernel after finite adaptation, supplies H3 with exponent $\upsilon = 1$. H4 is the following integrated local-oscillation condition for the actual fixed-preconditioner increment $\mathcal{H}_\vartheta$: for every compact $K$ and $\delta > 0$,

$$\sup_{\vartheta \in K} \int \Pi_\vartheta(dx) \sup_{\substack{\vartheta' \in K \\ \|\vartheta' - \vartheta\| \le \delta}} \|\mathcal{H}_{\vartheta'}(x) - \mathcal{H}_\vartheta(x)\| \le C_K \delta^\alpha, \qquad \alpha = 1. \tag{B5}$$

Smoothness of the mean field alone does not imply (B5); it is a stated regularity condition. Under H1–H4, the associated Poisson-equation solutions have the bounds used by Fort et al. Proposition 4.11. Their H6 requires a power-law gain exponent satisfying

$$\max\left\{\frac{1}{p}, \frac{1+(\alpha\wedge\upsilon)/p}{1+(\alpha\wedge\upsilon)}\right\} < a \leq 1. \tag{B6}$$

For $p \geq 2$ and $\alpha = \upsilon = 1$, the implemented $a = 0.80 > 0.75$ satisfies (B6), not merely the usual Robbins–Monro sums.

After safeguards become inactive, apply Fort et al. Proposition 4.6 to the stopped tail recursion with mean field $h = A\nabla\bar{\ell}$. Its C-i follows from differentiability, C-ii from compact-tail stability, and C-iii from the gain. For H5 and C-v(A), assume a coercive continuously differentiable extension $w$ that equals $-\bar{\ell} + C$ near the visited compact tail, has compact level sets and compact zero-drift set

$$\mathcal{L}_w = \{\vartheta : \langle\nabla w(\vartheta), h(\vartheta)\rangle = 0\},$$

such that $w(\mathcal{L}_w)$ has empty interior and $\mathcal{L}_w \cap K_{\text{tail}} = \mathcal{L} \cap K_{\text{tail}}$. This global extension and its critical-value property are assumptions, not automatic consequences of local likelihood smoothness.

For Proposition 4.6 C-iv, apply Proposition 4.11 and Corollary 4.12 to the process stopped on exit from the compact tail. For $A_k = \sup_{l\geq k} \|\sum_{j=k}^{l} \gamma_j \xi_j\|$, their result gives $P(A_k \geq \delta/2) \to 0$; because $A_n \leq 2A_k$ for $n \geq k$, continuity from above and countable rational $\delta > 0$ give $A_k \to 0$ almost surely on the eventual no-exit tail. Proposition 4.6 therefore yields $d(\vartheta_k, \mathcal{L}) \to 0$ almost surely. The finite prefix cannot change the limit set. This is a conditional stationary-set result for the Younes stochastic-gradient recursion presented as Delyon et al. equation (74), not an invocation of their finite-statistic SAEM Theorem 5 or 8.

For centered numerical scores, suppose $\|r_k(\vartheta, u)\| \leq h_k^2 M(u)$ with $\sup_k E\{M(Z_k)\} < \infty$. Taking $h_k = h_0\sqrt{\gamma_k}$ gives $E\sum_k \gamma_k \|r_k\| \leq Ch_0^2 \sum_k \gamma_k^2 < \infty$; hence the complete numerical perturbation is almost surely summable. Post-freeze one-sided differences are rejected because they lack this second-order bound.

# References


[1] Gunnar Yngman, Henrik Bjugard Nyberg, Joakim Nyberg, E. Niclas Jonsson, and Mats O. Karlsson. An introduction to the full random effects model. *CPT: Pharmacometrics & Systems Pharmacology*, 11:149–160, 2022. doi: 10.1002/psp4.12741.

[2] Joakim Nyberg, E. Niclas Jonsson, Mats O. Karlsson, and Jenny Haggstrom. Properties of the full random-effect modeling approach with missing covariate data. *Statistics in Medicine*, 43:935–952, 2024. doi: 10.1002/sim.9979.

[3] E. Niclas Jonsson and Joakim Nyberg. Full random effects models (FREM): A practical usage guide. *CPT: Pharmacometrics & Systems Pharmacology*, 13(8):1297–1308, 2024. doi: 10.1002/psp4.13190.

[4] Joakim Nyberg and E. Niclas Jonsson. The impact of misspecified covariate models on inclusion and omission bias when using fixed effects and full random effects models. *Journal of Pharmacokinetics and Pharmacodynamics*, 52(2):18, 2025. doi: 10.1007/s10928-025-09964-9.

[5] Abe Sklar. Fonctions de répartition à n dimensions et leurs marges. *Publications de l'Institut de Statistique de l'Université de Paris*, 8:229–231, 1959.

[6] Harry Joe. Families of m-variate distributions with given margins and m(m-1)/2 bivariate dependence parameters. In *Distributions with Fixed Marginals and Related Topics*, volume 28 of *IMS Lecture Notes–Monograph Series*, pages 120–141. 1996. doi: 10.1214/lnms/1215452614.

[7] Emmanuelle Comets, Audrey Lavenu, and Marc Lavielle. Parameter estimation in nonlinear mixed effect models using saemix, an R implementation of the SAEM algorithm. *Journal of Statistical Software*, 80(3):1–41, 2017. doi: 10.18637/jss.v080.i03. URL https://www.jstatsoft.org/article/view/v080i03.

[8] Radojka M. Savić, France Mentré, and Marc Lavielle. Implementation and evaluation of the SAEM algorithm for longitudinal ordered categorical data with an illustration in pharmacokinetics–pharmacodynamics. *The AAPS Journal*, 13(1):44–53, 2011. doi: 10.1208/s12248-010-9238-5.

[9] Bernard Delyon, Marc Lavielle, and Eric Moulines. Convergence of a stochastic approximation version of the EM algorithm. *The Annals of Statistics*, 27(1):94–128, 1999. doi: 10.1214/aos/1018031103.

[10] Estelle Kuhn and Marc Lavielle. Maximum likelihood estimation in nonlinear mixed effects models. *Computational Statistics & Data Analysis*, 49(4):1020–1038, 2005. doi: 10.1016/j.csda.2004.07.002.

[11] Vianney Debavelaere and Stéphanie Allassonnière. On the curved exponential family in the stochastic approximation expectation maximization algorithm. *ESAIM: Probability and Statistics*, 25:408–432, 2021. doi: 10.1051/ps/2021015. URL https://www.numdam.org/item/10.1051/ps/2021015/.

[12] Stéphanie Allassonnière, Estelle Kuhn, and Alain Trouvé. Construction of bayesian deformable models via a stochastic approximation algorithm: A convergence study. *Bernoulli*, 16(3):641–678, 2010. doi: 10.3150/09-BEJ229.

[13] Laurent Younes. Parametric inference for imperfectly observed gibbsian fields. *Probability Theory and Related Fields*, 82(4):625–645, 1989. doi: 10.1007/BF00341287.

[14] Estelle Kuhn and Marc Lavielle. Coupling a stochastic approximation version of EM with an MCMC procedure. *ESAIM: Probability and Statistics*, 8:115–131, 2004. doi: 10.1051/ps:2004007. URL https://www.numdam.org/item/10.1051/ps%3A2004007/.

[15] Gersende Fort, Eric Moulines, Amandine Schreck, and Matti Vihola. Convergence of markovian stochastic approximation with discontinuous dynamics. *SIAM Journal on Control and Optimization*, 54(2):866–893, 2016. doi: 10.1137/140962723.

[16] Samuel Gruffaz, Kyurae Kim, Alain Durmus, and Jacob Gardner. Stochastic approximation with biased MCMC for expectation maximization. In *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pages 2332–2340, 2024. URL https://proceedings.mlr.press/v238/gruffaz24a.html.

[17] Thomas A. Louis. Finding the observed information matrix when using the EM algorithm. *Journal of the Royal Statistical Society: Series B (Methodological)*, 44(2):226–233, 1982. doi: 10.1111/j.2517-6161.1982.tb01203.x.

[18] Christophe Andrieu, Éric Moulines, and Pierre Priouret. Stability of stochastic approximation under verifiable conditions. *SIAM Journal on Control and Optimization*, 44(1):283–312, 2005. doi: 10.1137/S0363012902417267.

[19] Ming Gao Gu and Fan Hui Kong. A stochastic approximation algorithm with markov chain monte-carlo method for incomplete data estimation problems. *Proceedings of the National Academy of Sciences of the United States of America*, 95(13):7270–7274, 1998. doi: 10.1073/pnas.95.13.7270. URL https://doi.org/10.1073/pnas.95.13.7270.

[20] Ming Gao Gu and Shaolin Li. A stochastic approximation algorithm for maximum-likelihood estimation with incomplete data. *Canadian Journal of Statistics*, 26(4):567–582, 1998. doi: 10.2307/3315718. URL https://doi.org/10.2307/3315718.